A-A-53470
22 August 1988

COMMERCIAL ITEM DESCRIPTION

GENERAL SPECIFICATION, BURS, BONE SURGERY, DIAMOND, FOR

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers general requirements for diamond, surgical burs used in various surgical procedures.

Salient characteristics:

Head shapes and shank lengths. The diamond surgical burs shall be of the following head shape and shank lengths as specified in the applicable item description sheets (See Notes).

Head shapes:

Round (Untapered Neck)
Round (Tapered Neck)
Oval
Rhinoplastic
Dermabrader

Shank lengths:

Medium
Long

General.

Commercial Item Description Sheet. The burs shall be in accordance with all requirements specified herein and the Commercial Item Description Sheet applicable for the head shape, head diameter, shank length and National Stock Number (NSN) (See Notes).

AMSC N/A FSC 6515

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Material.

Head. The abrasive shall be natural diamond bort. Diamond grit-size shall be #150 U.S. standard mesh.

Deposited density shall be 20 to 30 particles per square mm (0.0015 sq. in.) over spherical area of cutting surface. Diamond grit shall be bonded with nickel plating and shall not extend past the head.

Shank. The shank shall be fabricated from class 5, type 630 corrosion resistant steel in accordance with ASTM F899 and shall be properly hardened and passivated to a Rockwell hardness of C40 to C44.

Design and dimensions. The burs shall be diamond head, surgical burs with a round and non-notched proximal end. The bur shall consist of a one-piece shank, neck, and blank head to which the diamond particles are metal bonded. The design and dimensions of the burs shall be as specified in the applicable Commercial Item Description Sheet as specified herein (See Notes).

Runout. The total indicated runout of the diamond bur shall not exceed 0.005 inch when tested as specified herein.

Corrosion resistance. No more than 5 percent of the total area of the instrument shall show corrosion when tested as specified herein. Any such corrosion existing on diamond coated surfaces shall not exceed 2 percent of the total area of the instrument.

Finish. The neck and shank of the burs shall be smooth over the entire surface and any safeline present shall have an etched finish.

Plating. The mean value of exposed (unplated) and protruding height of diamond grit shall be 0.002 inch (0.05 mm).

Surface roughness. The surface roughness of the shanks of the burs shall be no more than the root-mean-square (RMS) average deviation of 20 micro-inches in the longitudinal direction of the shank, when determined as specified herein.

Identification marking. The shank of each bur shall be permanently and legibly stamped with the name or registered trademark of the manufacturer. Additional marking shall be at the option of the manufacturer.

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Instructions. A set of printed instructions outlining the use, care, sterilization, and method of cleaning diamond burs shall be enclosed with each unit package.

Workmanship. The diamond bur shall be symmetrically made and shall show no evidence of chipping, cracking or other defects which may detract from its appearance or impair its serviceability.

Unit. Each (EA). One bur, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of II. Examination shall include, but shall not be limited to the defects listed below. The unit of product for examination purposes shall be one bur.

Head shape, shank length or cutting length not as specified.

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Abrasive not uniformly distributed on head.
Finish of shank not as specified.
Identification marking not legibly, correctly, or completely stamped on shank, or missing altogether.
Laminations, cracks, bends, deformed portions, gouges, chips, or broken areas present in any part of bur.
Cutoff end of shank not as specified.
Safeline missing, not clearly defined or not of uniform thickness.
Shank edges and corners not smoothly chamfered or rounded.

Microscopic inspection. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 7 power (7X) magnification, unless otherwise specified herein.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified below. Unit of product for test purposes shall be one bur.

Characteristic	Inspection level	AQL (Percent defective)
Chemical analysis	*	**
Dimensions	S-2	2.5
Runout	S-1	1.5
Corrosion resistance	S-1	1.0
Surface roughness	S-1	1.0

*Sufficient amount from each lot of each metal used in the construction of the bur shall be selected for composition analysis. Tests shall be made on specimens representing each lot of diamond and the results averaged per lot.

**Acceptance number shall be zero for all sample sizes.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Chemical analysis. In the absence of certificates of quality for the raw material, the samples shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be use for referee analysis.

Dimensions test. Tests of dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator, sensitive to 0.0002 inch and equipped with a knife-edge foot, a shadowgraph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360 degrees rotation of the bur shall be measured and recorded to the nearest 0.0005 inch.

Corrosion resistance. The burs shall meet the requirements specified herein when subjected to the boil test as specified in ASTM F1089.

Surface roughness. The surface roughness of the shanks of the instruments shall be determined by using a Brush Surfindicator, or other equipment of equal accuracy. The RMS average shall be determined by squaring the height measurements of the peaks and valleys of the surface of the shank, obtaining the averages of the squared values, and then extracting the square root of this average square.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus resnipment from receiving activity, and shall conform to applicable carrier's rules and regulations.
Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Notes.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.

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(b) Title, number, and date of the applicable Commercial Item Description Sheet which shall include this specification letter and number, and the slash number. Examples:

A-A-53470/1 Bur, Bone Surgery, Diamond, 2 mm, Round Head.
A-A-53470/2 Bur, Bone Surgery, Diamond, 3 mm, Round Head.
A-A-53470/3 Bur, Bone Surgery, Diamond, 5 mm, Round Head.
A-A-53470/4 Bur, Bone Surgery, Diamond, 2 mm, Rhinoplastic Head.
A-A-53470/5 Bur, Bone Surgery, Diamond, 4 mm, Round Head.
A-A-53470/6 Bur, Bone Surgery, Diamond, 4 mm, Oval Head.
A-A-53470/7 Bur, Bone Surgery, Diamond, 4 mm, Dermabrader Head.

(c) Intermediate/exterior package quantities, labeling, and marking.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4389

51326

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NOTE This form may not be used to request copies of documents nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53470	GENERAL SPECIFICATION, BURS, BONE SURGERY, DIAMOND, FOR

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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